ECE 280 – Lab 1 Report
Signals and Systems: Arduino Signal Generator

Student Information

Course/Section: ______________________

Team Members: ______________________

Student IDs: ______________________

Lab Date: ______________________

TA/Instructor: ______________________

Submission Checklist

Submit this PDF report and supporting files (code + images) to your LMS.

- RC filter breadboard photo (labeled R, C, Arduino pin)
- DMM photos for 0%, 50%, 100% duty cycles
- Oscilloscope screenshot with time/div and volts/div visible
- MATLAB plot screenshot (500 samples, labeled axes)
- Arduino code and MATLAB script files
- Jitter comparison histograms (Phase 2 vs Phase 3)

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1 Pre-Lab Results (Summary)

Q1: Filter design target ($f_c \approx 50.000\,\text{Hz}$)

Chosen values: $R =$ ____________ Ω, $C =$ ____________ F

Computed cutoff: $f_c = \dfrac{1}{2\pi RC} =$ ____________ Hz

Q2: RC response

$h(t)$ summary and settling-time derivation ($\tau = RC$):

Q3: Quantization and SQNR

$\Delta V =$ ____________ V, SQNR (8-bit) = ____________ dB

2 Phase 1: Hardware Assembly and Verification

1.1 RC Filter Assembly

PWM pin used: ____________ Measured R: ____________ Ω Measured C: ____________ F

RC schematic used and wiring notes:

Breadboard photo:

Insert labeled breadboard photo here

Figure 1: RC filter hardware implementation.

1.2 Baseband Verification (DMM)

Table 1: PWM Duty-Cycle Verification

Duty Cycle	Expected (V)	Measured (V)	Percent Error (%)
0%	0.00		
50%	2.50		
100%	5.00		

Linearity comment (is output proportional to duty cycle?):

3 Phase 2: Acquisition and MATLAB Integration

2.1 Waveform Generation

Assigned waveform: ____________ Assigned frequency f_0: ____________ Hz

Implementation method (LUT/algorithmic/hybrid) and timing strategy:

2.2 Oscilloscope Capture

Table 2: Scope Measurements

Parameter	Value
Peak-to-Peak Voltage (V)	
Observed Frequency (Hz)	
Period (ms)	
Time/Div setting	
Volts/Div setting	
Visible distortion/ringing (Yes/No + note)	

Insert oscilloscope screenshot here

Figure 2: Oscilloscope capture of filtered waveform.

2.3 Serial Acquisition (MATLAB)

Serial port: ____________ Baud: 115200 Samples acquired: 500

Measured sampling period T_s: ____________ s Measured sampling frequency $f_s = 1/T_s$: ____________ Hz

Any acquisition issues or fixes:

2.4 MATLAB Visualization

Insert MATLAB time-series plot here

Figure 3: 500-sample MATLAB waveform plot with labeled axes and title.

4 Phase 3: Interrupt Optimization

3.1 Baseline Jitter (Delay-Based)

Table 3: Baseline Jitter Statistics (Phase 2)

Metric	Value (µs)
Mean interval	
Standard deviation	
Minimum interval	
Maximum interval	
Range	

3.2 Interrupt-Driven Implementation

Timer used: ________________ ISR frequency target: ________________ Hz

Brief ISR implementation summary and key register settings:

3.3 Jitter Comparison

Table 4: Jitter Improvement Summary

Metric	Phase 2 (Delay)	Phase 3 (Interrupt)
Mean interval (µs)		
Std. dev. (µs)		
Range (µs)		

Jitter improvement percentage:

$$\text{Improvement} = \frac{\sigma_{\text{phase2}} - \sigma_{\text{phase3}}}{\sigma_{\text{phase2}}} \times 100\% = \underline{\hspace{4cm}}$$

Insert Phase 2 vs Phase 3 jitter histograms here

Figure 4: Jitter distribution comparison.

5 Cross-Validation and Error Analysis

Table 5: Theory vs Measurement Comparison

Parameter	Theoretical	Scope	MATLAB	Error (%)
Amplitude (V)				
Frequency (Hz)				
Period (ms)				
Peak-to-peak (V)				

Primary error sources (rank top 3):

1.

2.

3.

Discussion: Explain mismatch between theory and experiments, and comment on interrupt impact.

6 Code Appendix

List submitted files and brief description.

File Name	Description

Academic Integrity Statement

We certify that this submission represents our own lab work and measurements, and that all submitted images/code correspond to our team's execution.

Signature(s): ____________________ Date: ____________________